Direction Grids for All-Characteristic Affine-Orbit Identity Testing

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Abstract

We refine the finite grids in the companion all-rank affine-orbit hitting theorem. At each rank stage, it suffices to preserve one minor for each of at most $(n+2)(n+3)/2$ distinct contiguous intervals. Digit polarization of a maximum-digit-weight leading monomial gives a nonzero tensor that is separately Frobenius-linear in every direction. Each interval test therefore excludes at most $n-1$ values per direction, independently of all degrees. The resulting bound is $Q^K(nD+1)(D+1)^K$, where $Q=(n-1)(n+2)(n+3)/2+1$ and $K=K(w)=2^{O(w^2)}$. Choosing globally between this argument and ordinary interpolation replaces Q by $\min\{Q,(n-1)D+1\}$ and restores the working-field requirement $|E|\geq nD+1$.

Status and dependency. The mathematical plaintext and this formal addendum have separate model-assisted internal reconstructions and exact-source audits. Their mathematical content is frozen after those internal reviews. This is not external peer review or a publication-priority claim. The companion manuscript supplies the all-characteristic symbolic rank-descent theorem and its Artin and Hasse lemmas. This addendum proves the finite-grid refinement and does not replace those structural proofs. The canonical companion is `manuscripts/general-affine.tex` in this problem directory. Its research-session source before the documented publication layout changes has SHA-256

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1 Model and quantitative statements

Let $n,w,D\geq 1$, let $\mathbb{F}$ be any field, and let the nonzero oracle polynomial have a representation

$$f(X)=uA_1(\ell_1(X)+c_1)\cdots A_n(\ell_n(X)+c_n)v,\qquad \deg f\leq D. \tag{1}$$

The univariate matrices A_i are w-by-w over $\mathbb{F}$, with arbitrary entry degrees and generic ranks. The forms ℓ_i are the coordinates of an invertible linear transformation. The order, affine transformation, and representation are unknown. The degree promise concerns only the scalar polynomial.

For $1\leq r<w$ put

$$W_r=\binom{w}{r},\qquad h_r=(r+2)\prod_{s=r+2}^{w}\left(\binom{s-1}{r-1}+1\right)-2, \tag{2}$$

$$k_r=(2h_r+2)(W_r^2-1),\qquad K=K(w)=\sum_{r=1}^{w-1}k_r. \tag{3}$$

Set $K(1)=0$. Empty products are one. The companion proves $K(w)=2^{O(w^2)}$ and that

$$f(\gamma(t)+\mathrm{SV}_K(z,y))\neq 0,\qquad \gamma_i(t)=t^i,$$

in every characteristic. Here $\lambda_1, \dots, \lambda_n$ are the Lagrange polynomials for n distinct nodes in a working field E extending $\mathbb{F}$, and

$$G_i(t, z, y) = t^i + \sum_{j=1}^{K} z_j \lambda_i(y_j).$$

Define

$$M = \frac{(n+2)(n+3)}{2}, \qquad Q = (n-1)M + 1. \tag{4}$$

These integers depend only on n, and $Q \le 2n^3$.

Theorem 1.1 (Degree-independent direction grids)**.** *Let $\mathcal{Y}, \mathcal{T}, \mathcal{Z} \subseteq E$ be arbitrary sets of respective sizes $Q, nD+1, D+1$. The evaluations*

$$\{G(t, z, y) : t \in \mathcal{T}, \quad z_j \in \mathcal{Z}, \quad y_j \in \mathcal{Y} \ (1 \le j \le K)\}$$

hit every nonzero polynomial (1). Their number is at most

$$Q^K (nD+1)(D+1)^K. \tag{5}$$

It suffices that $|E| \ge \max\{Q, nD+1\}$. The sets may overlap. For fixed w, the count is $O_w(n^{3K+1} D^{K+1})$.

The result also holds with the initial curve

$$\eta_i(t) = t\lambda_i(t) \tag{6}$$

in place of $\gamma_i(t)$. The corresponding evaluations lie in the image of pure SV_{K+1}, with its first pair specialized to $(z_0, y_0) = (t, t)$.

Corollary 1.2 (Smaller field and the better of two grids)**.** *Put*

$$B = \min\{Q, (n-1)D + 1\}.$$

Over a working field of size at least $nD+1$, a Cartesian hitting grid for either initial curve uses at most

$$(nD+1)\big((D+1)B\big)^K \tag{7}$$

queries. Every direction uses the same size-B set: the choice between the two arguments is made globally.

Extension-field evaluations require the corresponding black-box access. The statements do not promise that base-field-valued queries alone always suffice. Nonzero constants need one query. If $n = 1$, ordinary univariate interpolation gives the sharper elementary count $D+1$, independently of the displayed general upper bounds.

2 A nonzero Frobenius direction tensor

For characteristic $p > 0$, let $\mathrm{wt}_p(a)$ be the sum of the base-p digits of a. The digit weight of X^α is $\sum_i \mathrm{wt}_p(\alpha_i)$. In characteristic zero use ordinary degree, so every digit copy below has size one.

Lemma 2.1. *Let K_0 be a coefficient field containing E, and let $H(X)$ be a nonzero ordinary homogeneous polynomial of degree $m > 0$ and digit degree at most k. There are $k_0 \le k$ powers*

$q_1, \dots, q_{k_0}$ *of the characteristic, or ones in characteristic zero, with* $\sum_j q_j = m$*, such that the coefficient of* $\prod_j z_j^{q_j}$ *in*

$$H\left(\sum_{j=1}^{k_0} z_j V_j\right)$$

is a nonzero tensor

$$T(V_1, \dots, V_{k_0}) = \sum_{i_1,\dots,i_{k_0}} b_{i_1,\dots,i_{k_0}} \prod_{j=1}^{k_0} V_{j,i_j}^{q_j}. \tag{8}$$

After substituting $V_{j,i} = \lambda_i(y_j)$*, the tensor remains nonzero. For each direction* y_j*, at most* $n-1$ *values in* E *can make it identically zero in all the other directions.*

Proof. Choose a nonzero monomial X^α of *maximum* digit weight k_0 among the monomials of H. List its digit copies as the q_j, with repetition according to each digit. Their sum is m.

In the Frobenius expansion of a source monomial, every one of the k_0 positive amplitude exponents in the selected coefficient must receive at least one digit copy. No source monomial has more than k_0 copies. Thus every contributor has exactly k_0 copies, with exactly one assigned to each amplitude. In particular, no carry within an amplitude produces a mixed direction monomial. This proves the separate Frobenius-linearity in (8).

Assign the chosen monomial's copies to their corresponding direction coordinates. The resulting direction-monomial coefficient is the nonzero coefficient of X^α times

$$\prod_{i,e} \alpha_{i,e}!, \qquad \alpha_i = \sum_e \alpha_{i,e} p^e$$

in positive characteristic. Every digit is less than p, so all these factorials are nonzero. In characteristic zero the coefficient is instead $\prod_i \alpha_i!$, also nonzero. No division by $m!$ is used. A different source monomial cannot yield this same direction monomial, since its exponents sum back to α. Hence $T \neq 0$.

The functions $\lambda_i(y_j)^{q_j}$ are linearly independent over K_0. Their coefficient matrix on the powers $y_j^{aq_j}$ is the q_j-th Frobenius power of the invertible Lagrange coefficient matrix. In characteristic zero it is that matrix itself. Applying these injective linear maps in each tensor factor preserves $T \neq 0$.

View the resulting tensor as a polynomial in y_j over the rational function field in all the other directions. It is a nonzero linear combination of $\lambda_i(y_j)^{q_j}$. Over an algebraic closure of that coefficient field it is a q_j-th power of a polynomial of degree at most $n-1$. It therefore has at most $n-1$ distinct roots. This proves the assertion about values making the entire remaining tensor zero. Root extraction is only a proof device; neither perfection nor an algorithm for taking these roots is assumed. □

The maximum-digit-weight choice is necessary for this argument. Selecting an arbitrary lower-weight monomial would allow additional digit copies from other monomials to carry inside one amplitude and would not establish the special tensor form (8).

Lemma 2.2 (A finite family of tests)**.** *Suppose* M_0 *nonzero scalar tests over* K_0 *have first homogeneous components of digit degree at most* k*. Then one can specialize all* k *fresh SV directions to any prescribed set of* $(n-1)M_0 + 1$ *distinct elements of* E *so that every test remains nonzero as a function of the old parameters and the fresh amplitudes.*

Proof. A test with nonzero constant term imposes no condition. For every other test choose the nonzero tensor of Lemma 2.1, using at most k directions; additional directions are ignored. Specialize directions one at a time. As long as a tensor is nonzero, it excludes at most $n-1$

choices for the next direction. The union of the forbidden choices has size at most $(n-1)M_0$, so one of the prescribed values keeps all tensors nonzero. Repeat.

The chosen tensor is an amplitude coefficient of the original test. Higher ordinary homogeneous terms have larger total amplitude degree and cannot change it. Thus each test survives. This is an existential proof for an explicit Cartesian grid; the algorithm does not identify tests, tensors, or the good specialization. □

3 Counting the rank-descent conditions

We recall exactly the structural facts from the companion proof that are used here. It adjoins two constant scalar-endpoint factors to the n original matrices. At rank stage r, every augmented interval of generic rank greater than r already retains its rank at the current symbolic shift. A generic rank-r target interval is cut into shortest rank-r blocks and a higher-rank tail:

$$I = B_1 \cdots B_q T, \qquad B_j = C_j A_{i_j}.$$

Each C_j and T is empty or has every interval of rank greater than r. The required rank-r tests are

$$B_j, \qquad B_j B_{j+1} \ (j < q), \qquad B_q T. \tag{9}$$

Each is a contiguous interval of the same augmented word.

The companion's Artin-box and filtered local-Hasse arguments prove that *every nonzero r-minor* of any word in (9) has its entire first homogeneous residual component of digit degree at most k_r from (3). This is the filtered conclusion that permits Lemma 2.2; a low-weight witness somewhere in the series would not suffice.

There are $N = n + 2$ factors in the augmented word, hence at most $N(N+1)/2 = M$ nonempty contiguous intervals. For every generic rank-r target take its required intervals (9), and deduplicate the entire list by interval. Choose *one* nonzero r-minor of each distinct required interval.

Preserving that one minor certifies rank at least r; rank at most r follows from the identically vanishing larger minors. The chosen minor works in every target decomposition where the interval occurs. For its leading-component bound, choose any one such decomposition. There is therefore no extra factor for the number of minors, the number of target occurrences, or repeated decompositions.

At most M scalar conditions need to be preserved at each stage. Lemma 2.2 applies with the same grid size $Q = (n-1)M + 1$ for every one of the stage's k_r directions.

4 Fixing directions while retaining the rank invariant

This specialization must be done within the symbolic descent, not merely after a rational proof has been frozen. At the start of a stage, fix every earlier direction to its previously selected element of E, while keeping t and all earlier amplitudes algebraically independent. The current coefficient field is

$$K_0 = E(t, \text{earlier amplitudes}).$$

Every latent shift has the form

$$C_i = c_i + \ell_i(t, t^2, \ldots, t^n) + \text{a polynomial in earlier amplitudes}.$$

Its t-part is nonconstant, since ℓ_i is a nonzero linear form and $t, t^2, \ldots, t^n$ are independent polynomials over E. Thus each C_i remains transcendental over the original coefficient field and

over E, in every characteristic. This is the precise local-Hasse premise; no derivative condition is needed.

By induction all already prepared higher-rank interval minors survive as functions of the formal old parameters. The common direction grid preserves every new rank-r test as a function of these parameters and the new amplitudes. Old nonzero minors remain nonzero because setting new amplitudes to zero recovers them. The companion's adjacent-block chain argument consequently preserves every rank-r target after the directions have been fixed.

All unit pivots, coefficient-space bases, and constant cutwise gauges may be chosen anew over the current coefficient field. They are proof devices and introduce no additional direction conditions. In particular, we do not require an earlier rational expression for a gauge to retain its denominator under a later specialization. The actual tests are minors of original polynomial matrix intervals. Their analysis uses coefficients in the current old-parameter field, not new-direction denominators.

Iterating through the ranks gives at least one full choice of all K directions from the common set $\mathcal{Y}$ for which $f(G)$ is a nonzero polynomial in $t, z_1, \dots, z_K$. Its individual degrees are at most nD in t and D in each amplitude. The final $\mathcal{T}, \mathcal{Z}$ grids therefore hit it. This proves Theorem 1.1. Auxiliary representation or minor degrees do not enter this last interpolation step.

5 The alternative curve and the minimum-size rule

For (6), every nonzero latent linear combination is $c+tq(t)$, with q a nonzero combination of the independent Lagrange polynomials. It is nonconstant in every characteristic, and η has degree at most n. The entire stagewise argument and final degree counts are unchanged. This proves the pure-map specialization stated after Theorem 1.1.

Proof of Corollary 1.2. If $Q \le (n-1)D+1$, use Theorem 1.1 for all directions. Then $Q \le nD+1$, so its field requirement is simply $|E| \ge nD+1$. If $Q > (n-1)D+1$, use instead ordinary Cartesian interpolation on the full nonzero substituted scalar polynomial from the companion theorem. Every direction has degree at most $(n-1)D$, every amplitude degree at most D, and the initial parameter degree at most nD. This gives the stated size with $B = (n-1)D+1$.

The choice is global: one uses the same size-B direction grid for all pairs, justified entirely by one of these two arguments. No unproved mixture of stagewise degree and root-count bounds is needed. Both cases yield (7) over the advertised field. □

For fixed width, the degree-independent branch has exponent $K+1$ in the scalar degree instead of the basic $2K+1$ bound. Its dependence on n is larger. No optimality is asserted. The affine transformation must remain invertible; singular projections and unrestricted repeated variables are outside the model. All mathematical conclusions retain the internal-review status of the companion proof.